Semester Project: Intersection-based Indicators for Diagnostics of TPG's Network

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Abstract

This work is part of a larger project conducted by the Geneva Public Transport, TPG, which aims to define a supervised learning model to identify the points and causes that slow down public transport. Using AVL data and information on the signal transmitted by traffic lights, specific to the signaled intersections, this work aims to define a set of performance metrics that allow for analyzing the intersections, more specifically the traffic lights, in order to identify the most critical ones. The final set is composed of one explanatory variable and three diagnostic indicators. The performance metrics allowed for analyzing the traffic lights and identifying some of the possible main causes of public transport slowdown. Among these are the time lost at the traffic light waiting for the green light and possible structural limits of the intersection. By defining the metrics, it was possible to explore the AVL data for the intersections, recognize limitations, and identify possible improvements.

Contents

1 Introduction

In modern large cities, road congestion represents a structural challenge that impacts the functioning of entire urban networks. Beyond compromising private mobility, it generates chain effects on public transport (PT), causing systematic delays and undermining the perceived reliability of the service.

Commercial speed (VCOM), defined as the average travel speed of public transport vehicles from one stop to the next, is the crucial indicator of a transport network's health (Diab et al. 2012). However, its declines are often localized phenomena, directly linked to the performance of intersections. Congested or poorly coordinated crossings can generate systematic delays that propagate along entire lines, significantly degrading VCOM (Diab et al. 2012, De Keyser et al. 2018). While this parameter offers an overall view of the service's efficiency, its aggregate and retrospective nature limits its diagnostic value: VCOM measures the consequence of problems but does not identify their structural and localized causes. A significant methodological gap thus remains: the absence of a structured approach that allows for precisely locating the most critical intersections and segments and, at the same time, isolating the triggering factors; be they traffic signal planning, interferences with general traffic, excessive stop times, or infrastructural limits.

In the context of the city of Geneva, a commercial speed lower than the minimum value agreed between TPG and the Canton of Geneva, which should be 18 km/hr, is recorded. The poor performance, evaluated precisely through VCOM, is problematic both from the perspective of network users and from a budgetary standpoint; in fact, the failure to achieve certain performance objectives could lead to cuts in TPG's budget, which are reflected in a deterioration of the service offered. To limit these effects, it is crucial to have diagnostic tools capable of identifying not only where the PT network deteriorates but also why. This project is born precisely with the aim of bridging this methodological and operational gap. The guiding research question is: "How to identify, in a systematic and data-driven manner, the most problematic signaled intersections for public transport and, starting from these, diagnose the root causes of network criticalities?" The ultimate goal is to provide sector operators with an analytical tool that allows for targeted, efficient, and evidence-based interventions, transforming network data into concrete optimization actions.

2 Literature Review

The scientific literature concerning public transport and signalled intersection has evolved significantly thanks to the availability of high-resolution data. This review explores existing approaches for intersection optimization, the analysis of service improvement strategies, and the development of indicators for identifying criticalities in the network.

Regarding intersection optimization, several studies addressing the topic using a conventional approach are found in the literature. Traditionally, analysis has focused on improving performance for all vehicle types, seeking a compromise between private traffic and public transport. A significant example is the study by Shrestha et al. (2017), which analyzes the New Baneshwor intersection in Kathmandu. The research uses mixed indicators such as total vehicle volume, average delay, and queue length to evaluate the Level of Service (LOS). The primary goal in these cases is to model the existing situation and simulate alternative solutions (such as underpasses or different signal phases) to increase the overall capacity of the intersection. In a similar manner, De Keyser et al. (2018) examine the impact of Transit Signal Priority (TSP) on an intersection in Ghent. The study highlights an intrinsic tension: although TSP reduces travel times for trams, it can significantly degrade performance for other road users, especially under saturated conditions. The research suggests that absolute priority for PT is justified only if the number of on-board passengers exceeds a certain critical threshold, highlighting the need to balance the requirements of all stakeholders.

Beyond infrastructural optimization, various studies have examined public transport performance in the literature. Diab and El-Geneidy (2012) investigate the combined effect of different operational strategies by analyzing the impact of measures such as dedicated lanes, traffic signal priority, the use of articulated buses, and smart card payment systems. Using Automatic Vehicle Location (AVL) and Automatic Passenger Count (APC) data, the study demonstrates that the combined implementation of these strategies can reduce travel time by 10.5%. A crucial aspect that emerged is passenger perception:

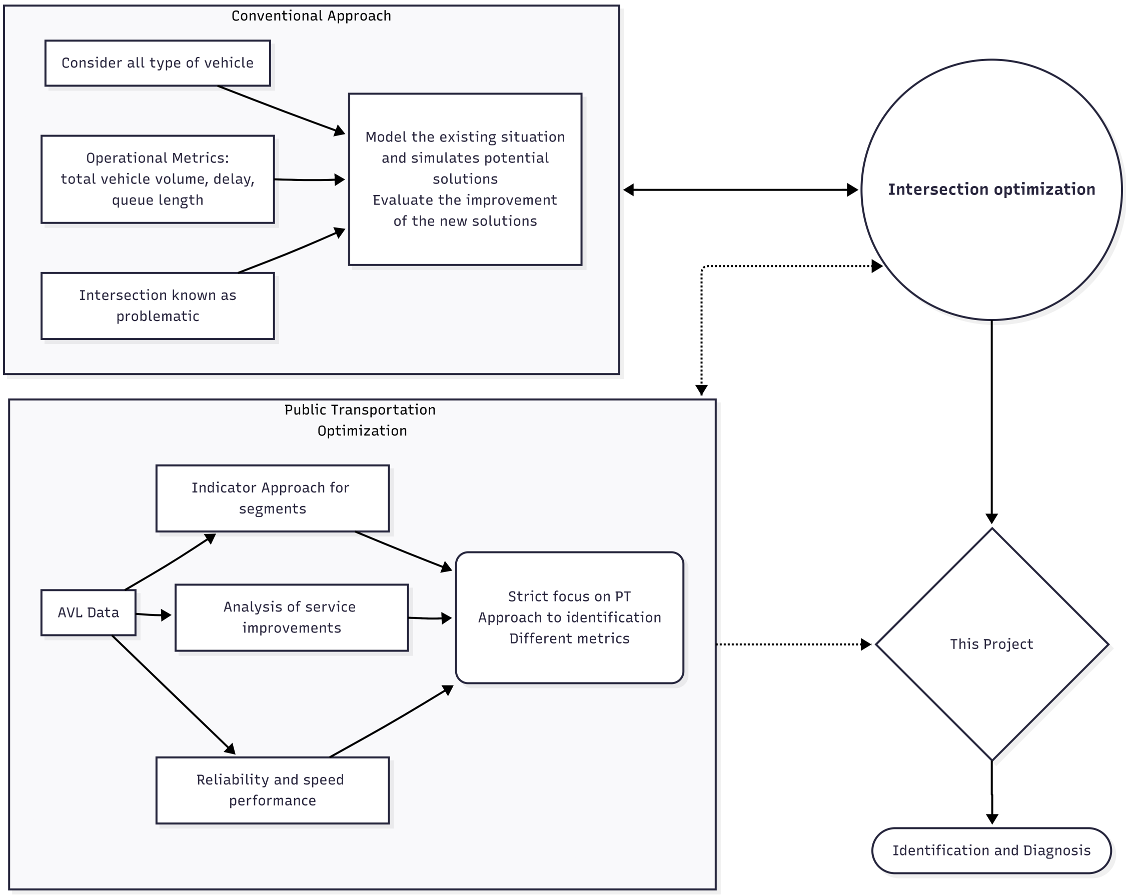

Figure 1: Summary diagram of existing literature

users tend to overestimate the actual time savings, showing a high degree of satisfaction with service improvements, even when technical data indicates modest savings. However, the research notes that some innovations, such as smart cards or articulated buses, can paradoxically increase dwell times if not supported by efficient boarding procedures.

A more recent line of research focuses on the systematic use of AVL data to automatically diagnose network problems. Brands et al. (2018) propose a method for identifying bottlenecks in a tram network in Amsterdam. They define eight criteria based on indicators such as: excessive dwell times, variations in punctuality, low cruising speed, and significant deviations from free-flow travel times. This approach shifts the focus from solving a known problem to the systematic diagnosis of the entire network, allowing for the localization of inefficiencies both spatially and temporally. However, the indicators are defined to analyze segments (stretch between two stops) and are not specific to intersections.

Parallelly, Hu and Shalaby (2017) explore service reliability at the route and segment level. Through a complex evaluation framework, they identify the coefficient of variation of travel times as the most appropriate measure for monitoring reliability. Their models reveal that the density of signaled intersections, the density of stops, and passenger volume are the factors that most influence service speed and regularity.

In this work:

From the analysis of previous works, a gap emerges: many studies analyze intersections in a general way, including private traffic, or focus on evaluating new strategies (prescriptive approach) rather than on problem identification (diagnostic approach).

This project fits into the framework of a larger project wanted by Geneva Public Transport, which aims to develop a model capable of identifying the most critical segments and intersections in the urban network, with the goal of proposing precise and data-driven interventions to optimize infrastructural investments. The main objective of this work, therefore, is to define a set of performance indicators to evaluate the impact of intersections on public transport (trams, buses). The study's approach for analyzing intersections distinguishes itself from the more traditional one in particular for two characteristics:

1. Exclusive focus on public transport: it does not account for the performance or impact of other private vehicles using the intersection.
2. Diagnostic, not prescriptive: the primary purpose is the identification and precise understanding of problems. This fundamental diagnosis is a crucial first step before developing specific solutions, ensuring that interventions are targeted and effective.

3 Methodology

3.1 Project Overview

The main project defined by TPG is broader and consists of three sub-projects that aim to define indicators allowing for the analysis of the Geneva urban network. The three sub-projects are specific for analyzing segments (stretches between two stops), signaled intersections, and predicting the travel time of segments. The ultimate goal is to combine the indicator sets into a set of features, which will constitute the input for a supervised learning framework designed to detect problematic areas of the network. This work focuses on intersections and, in particular, at the traffic light level. The analysis is done considering individually the traffic lights present at the intersection.

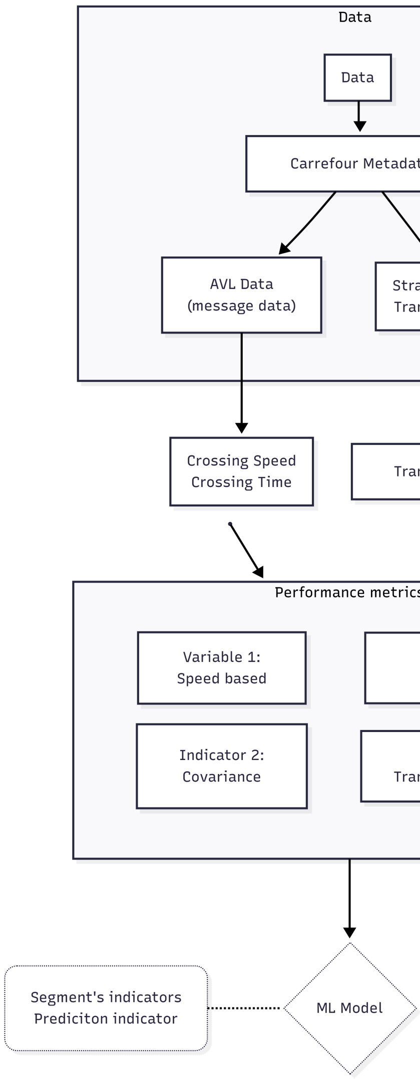

Figure 2: Overview of the project methodology and workflow

3.2 Data

The main available data is the automated vehicle location (AVL) data. TPG has over 10 years of data collected from vehicles. In this project, a reduced sample of data, one week, was used to have a sufficient number of crossings, observations, to define the indicators without requiring too much computation time. Besides the available AVL data, there is metadata for the intersections and the signal transmitted every second from a selection of strategic intersections during peak hours.

Intersection metadata

The first dataset contains the intersections that constitute the route of each line. For each line and direction, all intersections and the information necessary to use the AVL data are listed. The most relevant information used is summarized in Table 1

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Table 1: Description of the metadata for intersections

Variable	**Description**
Line ID	Identification number for each line
Intersection ID	Identification number for the intersection
Traffic light ID	Identification number of the traffic light. Generic IDs that repeat in the network. Associated with the Intersection ID it was used to build a specific ID for each traffic light (ex: `65 - V1`, indicates traffic light V1 at intersection 65)
Distance before/after	Distance of the receiver before/after the intersection. They were used to calculate the actual physical distance that vehicles travel
Tab before/after	Each vehicle sends different messages based on its position and the action performed. There are three types of messages: Tab0, Tab1 and Tab2. This information allows knowing the type of message corresponding to the moment when the vehicle arrives at and exits the intersection
No. reference balise before/after	Indicates the reference number of the receiver (balise) that receives/sends the message.

Automated Vehicle Location Data (message data)

The second dataset contains all AVL data recorded by the vehicles. Each recorded datum corresponds to a message, not all messages are relevant for defining a crossing. Each message contains the information summarized in Table 2. Based on the metadata for the intersections, it is possible to filter and select only the two relevant data points (before/after). With the recordings necessary to characterize the crossing, it is possible to calculate the crossing time and speed. To calculate the speed, it is necessary to determine the distance between the two recordings; being GPS data, it could be imprecise; using the actual distance determined with the metadata makes it possible to filter wrong measurements.

Table 2: Description of the information recorded for each message

Variable	**Description**
Crossing ID	Identification number for each event. All data recorded for a crossing have the same number. Allows isolating the two relevant messages.
Intersection ID, line ID, and traffic light ID	Identification numbers for intersection, line, and traffic light. Also in this case, for each traffic light, the ID was defined following the logic intersection - traffic light
Date and time	Indicate the exact moment when the datum was recorded
Distance	Indicates the position of the vehicle relative to the traffic light. Being GPS data, it may be imprecise compared to that in the metadata dataset
Tab	Indicates the type of message that was recorded
No reference balise	Indicates the receiver that recorded the message. Combined with the Tab it allows selecting the relevant message for the event (before or after the intersection)

Signal transmitted by the traffic light

This dataset is different from the other two, both in terms of the information contained and the space and time considered. The data is available only for about sixty intersections considered strategic, due to their location and distribution in the network, and is available only for the morning and evening peak hours of a single day (November 13, 2025). The morning peak hours are from 07:00 to 09:00 while the evening ones are from 16:00 to 18:00. For each traffic light at an intersection, the signal transmitted every second is indicated; the types of signals are distinguished by a different code (see Appendix A). Table 3 represents an excerpt of the processed dataset where each column represents a traffic light (intersection - traffic light).

Table 3: Extract from the dataset

Timestamp	`100 - BB`	`100 - B10`	`140 - V2`	`168 - V2`	`168 - B12`
2025-11-13 07:02:50	3	16	48	48	3
2025-11-13 07:02:51	3	16	48	48	3
2025-11-13 07:02:52	15	16	48	48	3
2025-11-13 07:02:53	15	16	48	48	3
2025-11-13 07:02:54	48	16	48	48	3
2025-11-13 07:02:55	48	3	48	48	3
2025-11-13 07:02:56	48	3	48	48	3
2025-11-13 07:02:57	48	3	12	48	3
2025-11-13 07:02:58	48	3	12	48	3
2025-11-13 07:02:59	48	3	12	12	3

3.3 Performance metrics

The available data allows obtaining the speed, the crossing time, and the traffic light signal when the message was recorded. Starting from these metrics, 4 performance metrics were defined: one explanatory variable and three diagnostic indicators.

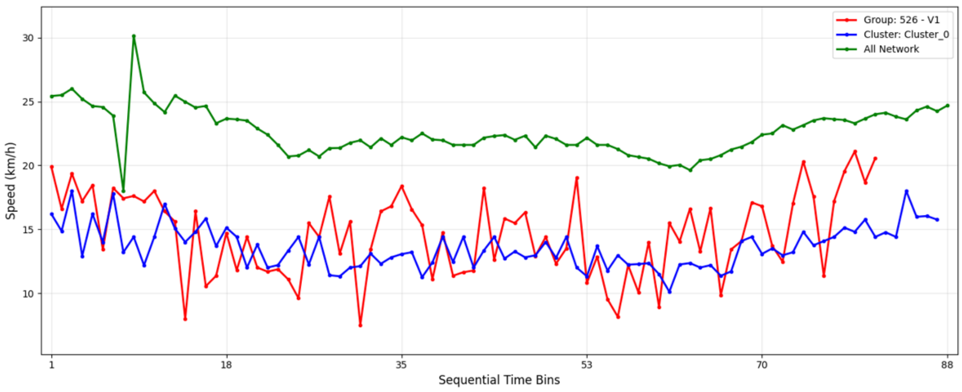


Figure 3: Example of traffic light `526 - V1` to show the advantages of the reference median speed based on clusters

3.3.1 Explanatory variable, V1 - median speed

The first metric evaluates the performance of the traffic light by comparing it with others that have similar characteristics. The metric used is the median crossing speed, defined for 15-minute intervals. This is compared with a reference median speed. Initially, as the reference median speed, $v_{50,\text{net}}$ was considered, i.e., the one considering all traffic lights in the network; however, it emerged that doing so did not account for the intersection length. Length is a parameter that has a particular influence on speed, especially in situations where the vehicle must stop at the traffic light and start again. In cases where the vehicle finds a clear passage or the priority system is activated and it is granted rapid clearance, the vehicle approaches the intersection at a high speed and continues without having to stop, at most slowing down slightly, and thus the speed remains high and does not depend on the length. On the other hand, in cases where the vehicle must slow down significantly or stop at the traffic light, the length has a greater influence. Once it receives clearance, the vehicle accelerates and crosses the intersection. In these cases, if the intersection is short, the vehicle might pass the recording point (after) before reaching the maximum speed (road limit) and the average crossing speed will be lower. In the cases of longer intersections, however, the vehicle has more meters to travel and thus will be able to reach higher speeds. Considering these two situations and assuming that vehicles have the same acceleration, the speed of the shorter intersection will be lower. To address this problem, each traffic light was assigned to a cluster based on length. The clusters were defined using the wcss and the elbow method. For each cluster k, $v_{50,k}$ was calculated, the reference median speed of the cluster. The indicator, therefore, compares for each 15-minute interval the median speed of each traffic light with $v_{50,k}$ of its cluster. The difference for each time interval is normalized by dividing by the maximum absolute value. The result will be between 0 and 1, where 0.5 represents a difference of zero.

The advantages of using length clusters

The graph in Figure 3 allows illustrating the advantage of using length clusters to define the reference median speed instead of the entire network. The green line corresponds to $v_{50,\text{net}}$, the blue one to $v_{50,0}$ (cluster 0 in this case) while the red one is for a single traffic light. Considering $v_{50,\text{net}}$, this traffic light always has a median speed lower than the reference and would thus be considered potentially problematic. If instead, $v_{50,k}$ is considered, the traffic light functions similarly to the reference; in some time intervals the speed is higher, in others lower. In general, the reference based on clusters allows for a better grasp of traffic light dynamics.

3.3.2 Indicator 1: Crossing time

The first diagnostic indicator is based on the intersection crossing time. A correctly functioning traffic light that favors public transport passage should guarantee short crossing times, relative to the time needed to cross the traffic light, and substantially uniform. This indicator is defined in two parts. The first consists of defining 3 clusters for each interval based on the crossing time for each 15-minute interval. The clusters are determined with the `k-means` algorithm and are ordered according to the increasing order of the centroids; thus cluster 1 represents the one with the shortest time, while cluster 3 the one with the longest time. The centroids of each cluster represent the average value of the observations in the cluster. Subsequently, according to equation 1, a score was defined. The score is based on the proportions of each cluster and they are multiplied by a weight. The weights for clusters 1 and 3 are fixed, respectively $w_1 = 1$ and $w_3 = 0$. The values are dictated by the desire to give more importance to observations with a short crossing time. Regarding the weight of cluster 2, w_2, the weight varies for each traffic light and time interval and was calculated according to equation 2. The idea is that the closer the crossing times in cluster 2 are to those in cluster 1, the higher the weight. In this way, the complete distribution of times within cluster 2 is considered, normalizing them relative to the interval defined by the extreme clusters (clusters 1 and 3). This allows for capturing the traffic light's behavior more accurately: if vehicles in cluster 2 generally have short times (close to cluster 1), the weight will be high; if instead the times tend towards long values (close to cluster 3), the weight will be low. Using a weight calculated on the normalized average of all observations in cluster 2, the distribution of crossing times is taken into account more comprehensively, rewarding traffic lights that present short times not only in cluster 1 but also in a significant portion of cluster 2. The result will be between 0 and 1, where one represents good functioning.

$$I_{1,i} = \sum_{j=1}^{3} C_{ij} \times w_{ij} \tag{1}$$

$$w_{i1} = 1, \quad w_{i2} = 1 - \frac{1}{n_{i2}} \sum_{k=1}^{n_{i2}} \frac{t_{i2,k} - c_{i1}}{c_{i3} - c_{i1}} = 1 - \bar{n}i2, \quad w_{i3} = 0 \tag{2}$$

where:

- i: indicates a single traffic light
- j: indicates the cluster, $j = 1, 2, 3$
- C_{ij}: proportion for traffic light i in cluster j
- w_{ij}: weight for traffic light i in cluster j
- c_{ij}: is the centroid for traffic light i in cluster j
- $t_{i2,k}$: k-th crossing time in cluster 2 for traffic light i
- n_{i2}: number of observations in cluster 2 for traffic light i
- $\bar{n}_{i2}$: normalized average of the times in cluster 2, defined as:

3.3.3 Indicator 2: Coefficient of variation CV

The second diagnostic indicator seeks to measure the reliability of crossing performance using the coefficient of variation, C_v. This choice is based on the result obtained from the study by Hu and Shalaby (2017) where the coefficient of variation of total travel time proved to be the most suitable metric. Dealing only with intersections and not the entire route, in this work it was calculated considering speed, $I_{2\mathrm{s}}$, and crossing time, $I_{2\mathrm{t}}$.

$$I_{2\mathrm{s},i} = \frac{\sigma_v}{\mu_v} \quad \text{and} \quad I_{2\mathrm{t},i} = \frac{\sigma_t}{\mu_t} \tag{3}$$

where σ represents the standard deviation and μ the mean of the speed (s) and crossing time (t).

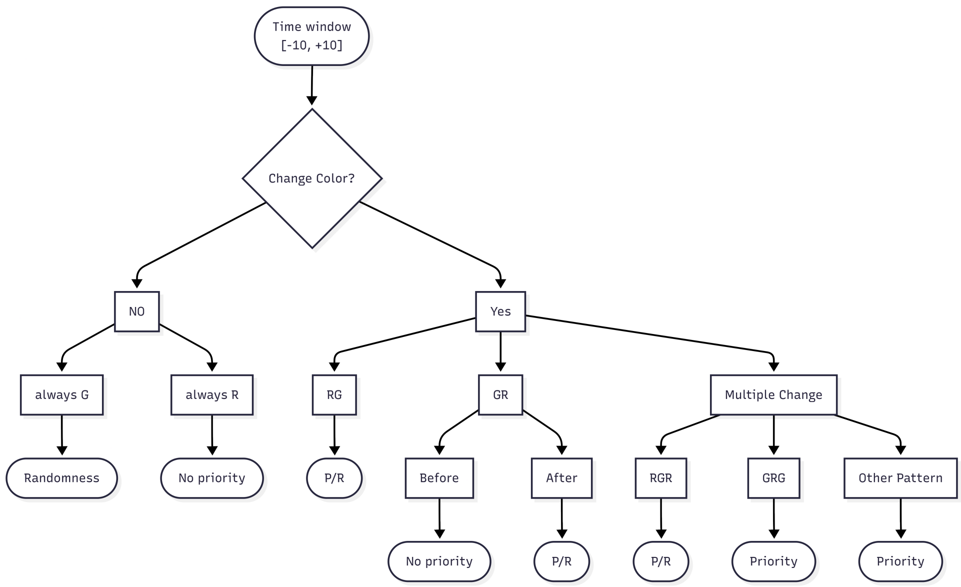


Figure 4: Decision tree for classifying each message into: coincidence, priority, and non priority

3.3.4 Indicator 3: Transmitted signal

The fourth performance metric differs from the others because it combines AVL data with transmitted signal data. The goal is to analyze how the traffic light behaves when a vehicle approaches the intersection. For this indicator, the pair of messages characterizing a crossing was not considered, but only the message sent before the intersection. Based on the exact time of the recording, obtained from AVL data, it was possible to obtain the signal transmitted by the traffic light. Since the measurement point does not correspond with the traffic light line, a time window of $\pm X$ seconds before and after the message recording was considered (as an initial size, $X = 10$ was taken). The time window aims to analyze how the transmitted signal changes as the vehicle approaches through patterns and, according to the decision tree in Figure 4, assign each message to one of the following categories: *Randomness*, *Priority* and *Non Priority*.

Managing uncertainty due to coincidence

Coincidence refers to situations for which there is insufficient information to determine whether the traffic light identified a vehicle and gave priority to public transport or whether the vehicle had the green light by coincidence. All time windows in which the signal does not change and is always green are categorized as *Randomness* because they correspond to the situation where the vehicle arrives in the middle of a green phase. The condition of uncertainty is also present in time windows with a pattern suggesting vehicle recognition and grant of clearance. An example is the situation that changes from red to green; in this case, with the arrival of the bus, the traffic light turned green, granting clearance; however, there is no certainty that the system gave the green light to public transport or, again, a coincidence. These cases are defined as *Priority or Randomness* but are ultimately considered in the *Priority* category. The explanation of other possible patterns is found in Appendix B.

Definition of the indicator

With the goal of reducing the uncertainty caused by coincidence, for defining this indicator only the *Non Priority* category was considered, because according to the decision tree it is the only category for which there is certainty that if the vehicle was not recognized, it had to stop at the traffic light. In this case, normalization is not necessary because the value is already between 0 and 1, where 0 means no vehicle had to stop at the traffic light, while 1 means the opposite.

4 Result

In this section we will attempt to delve deeper into the outputs of the different performance metrics. First, a correlation analysis was conducted to study the dependencies between the indicators, and subsequently, we tried to interpret the outputs and provide a preliminary diagnosis of the possible causes of public transport delays. For I_3, the influence of the time window size on the patterns was also evaluated.

4.1 Correlation Analysis

The conducted correlation analysis uses Pearson's correlation coefficient to evaluate the linear relationship between pairs of variables. Initially, the analysis was performed considering the first three performance metrics and different time periods. Subsequently, the average value during peak hours was calculated, and the correlation analysis was performed also considering I_3.

The first correlation analysis was conducted on a total of 60,389 observations. All correlations are statistically significant, and the results, summarized in Table 4, reveal the following main trends. A notable consistency in the sign and magnitude of the correlations across all analyzed time periods is observed. Specifically, the relationship between $\mathrm{I}_1 \leftrightarrow \mathrm{I}_{2t}$ shows a moderate to strong positive correlation. The indicators based on the coefficient of variation, $\mathrm{I}_{2t} \leftrightarrow \mathrm{I}_{2s}$, show a strong and very stable positive correlation between them. The correlation between $\mathrm{V}_1 \leftrightarrow \mathrm{I}_1$ is positive but weak, while the relationships between $\mathrm{V}_1 \leftrightarrow \mathrm{I}_2$ are instead characterized by weak negative correlations.

Table 4: Correlation matrices for different time periods. (* : $p < 0.05$)

	All Day				**Weekday**				**Weekend**		
	V_1	I_1	I_{2t}		V_1	I_1	I_{2t}		V_1	I_1	I_{2t}
I_1	0.263*	–	–	I_1	0.262*	–	–	I_1	0.183*	–	–
I_{2t}	-0.168*	0.391*	–	I_{2t}	-0.155*	0.405*	–	I_{2t}	-0.180*	0.335*	–
I_{2s}	-0.244*	0.132*	0.611*	I_{2s}	-0.213*	0.156*	0.643*	I_{2s}	-0.307*	0.099*	0.643*

	Morning Peak				**Midday**				**Evening Peak**		
	V_1	I_1	I_{2t}		V_1	I_1	I_{2t}		V_1	I_1	I_{2t}
I_1	0.317*	–	–	I_1	0.302*	–	–	I_1	0.334*	–	–
I_{2t}	-0.150*	0.361*	–	I_{2t}	-0.155*	0.378*	–	I_{2t}	-0.11*	0.4*	–
I_{2s}	-0.244*	0.09*	0.603*	I_{2s}	-0.232*	0.107*	0.611*	I_{2s}	-0.203*	0.123*	0.583*

The second correlation analysis was conducted on a total of 55 observations per indicator, one value for each strategic traffic light. The analysis shows a moderate negative correlation between $\mathrm{I}_3 \leftrightarrow \mathrm{V}_1$ and a moderate to strong negative correlation between $\mathrm{I}_3 \leftrightarrow \mathrm{I}_1$. These results suggest a relationship between the metrics, which will be considered in the following paragraphs where we will attempt to interpret the output of each metric. Regarding the relationship with the coefficients of variation, I_{2t} and I_{2s}, the correlation is weak and only in one case statistically significant.

Table 5: Correlation matrices considering the average values of peak hours. (*: $p < 0.05$)

	Morning Peak					**Evening Peak**			
	V_1	I_1	I_{2t}	I_{2s}		V_1	I_1	I_{2t}	I_{2s}
I_3	-0.377*	-0.449*	0.11	0.299*	I_3	-0.374*	-0.425*	0.016	0.215

4.2 Influence of the Time Window Size

Indicator 3 is defined considering a time window; the idea of the window is to consider the period of time during which the traffic light recognizes the arrival of public transport, sends the signal to the

Table 6: Proportions of the categories for different time windows - Mean ± Std.Dev

Category	**TW ±8**	**TW ±10**	**TW ±15**
priority	0.28 ± 0.17	0.36 ± 0.20	0.51 ± 0.23
randomness	0.16 ± 0.19	0.11 ± 0.16	0.05 ± 0.10
non-priority	0.56 ± 0.26	0.51 ± 0.26	0.44 ± 0.25

system which must process it and grant free passage. The size can depend on how the public transport priority system has been set up or also on the needs and standards of the TPG. To see how the time window size can influence I_3, Table 6 summarizes the mean and standard deviation of the categories based on time windows of ±8, ±10, ±15 seconds. Observing the mean value among the traffic lights, as the size increases the proportion of the *Non Priority* and *Randomness* categories decreases while that of the *Priority* category increases. The standard deviation, however, remains constant, suggesting that the size influences all traffic lights in a similar way. These observations were expected and are not surprising; indeed, a longer time window increases the chances of having a signal change and a pattern suggesting possible priority, and reduces the cases of *Non Priority*.

4.3 Interpretation

The four performance metrics (see 3.3.1, 3.3.2, 3.3.3, 3.3.4) attempt to analyze intersections using data and metrics that capture different trends. The explanatory variable, V_1, allows for a quick understanding of the traffic light's performance compared to similar ones. Values tending towards the extremes, 0 or 1, suggest that the traffic light is generally fast ($V_1 \approx 1$) or slow ($V_1 \approx 0$). A less clear behavior occurs when $V_1 \approx 0.5$. In this case, the traffic light could have a constant speed around the reference value or high variability due to some fast crossings and other slow ones. The case of constant speed suggests rather good operation, while the other suggests random operation, with fast or slow crossings without a consistent pattern.

Indicator I_1 was defined considering the crossing time; cluster 1 (short) represents approximately the minimum time needed to cross the intersection. A high value ($I_1 \approx 1$) indicates that most observations pass the intersection in a short time. Conversely, a low value indicates that most vehicles cross the intersection in a time longer than necessary, and thus the cause of the slowdown is due to a loss of time. The main cause of this loss could be waiting at the traffic light for the green signal, but it could also be due to other reasons such as the presence of pedestrians or traffic within the traffic light zone. Indicator 1 does not allow for a precise definition of the cause of the time loss.

The interpretation of I_2 distinguishes itself from the other two because it allows evaluating traffic lights from the point of view of performance consistency. A high value ($I_2 \approx 1$) allows identifying traffic lights with irregular performance, suggesting that public transport vehicles are considered at the same level as private vehicles, they share lanes and depend on traffic volume and flow. This indicator allows adding information to the interpretation of the other indicators. For example, revisiting the case where $V_1 \approx 0.5$, if $I_2 \approx 0$ it suggests that the traffic light has a roughly constant speed around the reference one. Conversely, if $I_2 \approx 1$, it is likely that the traffic light is characterized by irregular performance, with some fast crossings and other slow ones. Another interesting situation is when a good value for I_2 is accompanied by a low value of V_1. In this case, the reduction in speed could be caused by structural limits of the intersection.

Finally, I_3, allows a more specific interpretation compared to I_1. A high value indicates that vehicles must stop at the traffic light and wait for free passage. This represents one of the causes that slow down buses, namely the loss of time due to waiting at the traffic light. This observation is reinforced if we consider the correlation results between $\mathrm{I}_3 \leftrightarrow \mathrm{V}_1$ and I_1. The moderate to strong negative correlation confirms that the main cause of time losses is waiting for the green light at the traffic light and as a consequence there is a slowdown of public transport.

Table 7: Comparison between timebin 07:00-07:15 values and average peak hour values

	Timebin 07:00 - 07:15			**Average peak hour value**		
	`68 - V7`	`21 - V3`	`168 - B12`	`68 - V7`	`21 - V3`	`168 - B12`
V_1	0.25	0.69	0.5	0.2	0.68	0.42
I_1	0.6	0.9	0.8	0.57	0.89	0.68
I_{2t}	0.63	0.30	0.72	0.56	0.62	0.58
I_{2s}	0.67	0.13	0.51	0.67	0.16	0.52
I_3	–	–	–	0.67	0.08	0.45

Examples with some traffic lights:

To interpret the performance metrics together and try to understand possible relationships, three traffic lights were selected and used as an example. The three traffic lights were selected because they perhaps represent the three most classic and simplest cases to define; however, within the entire network, many other very different cases can be found. Table 7 contains both the values of a single timebin and the average values for peak hours; in general, the average values correspond to those of the timebin, only I_2, in some cases, shows substantial differences between the values suggesting different dynamics in the different timebins. The first example, traffic light `68 - V7`, represents problematic behavior. The median speed is lower than the reference ($V_1 = 0.2$), indicating that the traffic light is slow. The values of the other indicators indicate that PT vehicles lose time due to waiting at the traffic light (according to I_1 and I_3). A completely opposite example is represented by traffic light `21 - V3`, which generally has a high speed, superior to the reference one. The values of I_1, I_{2s} and I_3 suggest that this traffic light has a functioning recognition and priority system for PT vehicles that allows for good performance. Only I_{2t} shows a high value, suggesting much variation in crossing time, a result difficult to interpret. Finally, traffic light `168 - B12`, represents a more complex case. It has a speed similar to the reference one, slightly slower, while the other 3 indicators suggest that the traffic light has a behavior tending towards problematic but in a different way compared to the first example. In particular, this traffic light might be characterized by some fast crossings and other slow ones; this results from $V_1 \approx 0.5$ accompanied by a moderately high I_1, which suggests that in some crossings the PT vehicles lost time, while the coefficients of variation suggest moderately irregular behavior. The cause of the irregular behavior could be due to the fact that in about half of the observations the vehicles had to stop at the traffic light ($I_3 = 0.45$).

5 Discussion

Strengths

Among these, an interesting aspect is the discretization of the day into 15-minute intervals. This allows for a more detailed analysis of trends that may vary depending on different periods; at the same time, it allows considering larger time bands by aggregating the values of the different intervals. Another interesting point is the fact that the performance metrics use a diversified methodological approach, analyzing traffic lights considering different levels and metrics. In particular, the explanatory variable stands out because it studies the traffic light from an external point of view, comparing it with other traffic lights. The other metrics, instead, are specific to each traffic light, but use different metrics and data. This allows for an analysis that considers multiple factors, such as physical performance, service reliability, and traffic light behavior, favoring a more global interpretation that allows for a preliminary diagnosis of the causes that slow down public transport.

Limitations

The limitations of the performance metrics can be summarized into two main groups: limitations of the AVL data and excessive generalization. The AVL data represent a limitation because they allow defining all traffic lights uniquely with the two recording points. This limits the description of the intersections; in particular, they are used to approximate the start and end point of each reported crossing without taking into consideration possible elements that might be present from the recording point to the traffic light line that characterize each intersection and could be a cause of slowdown. Being the only two points, they were also used to approximate the point where the priority system could be activated. Beyond this, they do not allow obtaining information about what happens within the traffic light zone; this could be useful for identifying other possible causes of slowdown. The other group of limitations stems from excessive generalization; by this, we mean the fact that no distinction was made between vehicles when in reality trams, buses, articulated or electric buses can have different mechanical characteristics and planning hierarchies that could allow better explanation of some observations. The intersections were also all approximated to a straight line without considering curves, changes of direction, and origin from different roads. Finally, the dependence of I_3 on the window size could constitute a limitation, especially if the system details are not known or if each traffic light is set up differently.

Possible improvements

The excessive generalization is due to a lack of information that does not allow characterizing each traffic light in depth. Unlike the AVL data, which provide only two points, this information could be obtained from the Canton of Geneva or from TPG themselves. In particular, we mean information regarding the type of vehicle, dedicated bus lanes, the number and duration of the traffic light phases, the hierarchy of the origin and destination road, but also information regarding traffic and weather. This information would not allow defining other performance metrics but would allow better contextualizing each traffic light and identifying different dynamics and modes of operation. The advantages of having more information on the characteristics of traffic lights are demonstrated by the fact that, considering a reference speed based on the length of the intersection, it was possible to observe dynamics that would not have emerged if the entire network had been used as a reference. It must be noted that the data were partly used in a superficial manner and were not fully exploited. For example, the dataset with the signal data transmitted every second from the traffic light could be used to determine whether traffic lights have constant or irregular cycles and allow better management of the problem due to randomness when analyzing the patterns of the time windows.

6 Conclusion

The primary objective of identifying the traffic lights that cause the greatest slowdowns to TPG vehicles was pursued through an approach that distinguishes itself from conventional literature due to its exclusively diagnostic nature and focus on public transport. Beyond the four metrics, initially, this work allowed for exploring the AVL data on the reported intersections. In particular, it allowed understanding their structure and the way to manipulate them to obtain the necessary information for each crossing. The in-depth study of these data allowed defining the set of indicators but also brought to light some limitations in the analysis due mainly to the granularity of the data. Subsequently, four performance metrics were defined: one explanatory variable and three diagnostic indicators. The explanatory variable allows for a quick understanding of each traffic light by comparing it with others that have similar characteristics, specifically the length, while the diagnostic indicators allow analyzing service reliability and the capacity to give priority to public transport and determine the causes of service slowdown. While I_1 and I_3 highlight time losses due to waiting for the green light or physical obstacles, indicator I_2 provides a crucial measure of service reliability, revealing whether public transport performance is excessively dependent on fluctuations in private traffic. By defining the metrics, in addition to some limitations of the AVL data, the need emerged to include more information regarding the intersections, which can allow for a greater understanding of the outputs and the reasons for slowdown. Despite these observations, it must be recognized that the set of performance metrics allows capturing the dynamics of the traffic lights and can represent a starting point for an even more in-depth analysis of the intersections.

References

Brands, Ties, Niels van Oort, and Menno Yap (2018). "Automatic bottleneck detection using AVL data: Caspt 2018: 14th Conference on Advanced Systems in Public Transport and TransitData 2018". In: *Proceedings of Conference on Advanced Systems in Public Transport (CASPT) 2018*.

De Keyser, Olivier et al. (Oct. 2018). "Optimising the public transport priority at road intersections". en. In: *IET Intelligent Transport Systems* 12.8, pp. 986–994. ISSN: 1751-956X, 1751-9578. DOI: 10.1049/iet-its.2018.5133. URL: https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/iet-its.2018.5133 (visited on 01/07/2026).

Diab, Ehab I. and Ahmed M. El-Geneidy (Mar. 2012). "Understanding the impacts of a combination of service improvement strategies on bus running time and passenger's perception". en. In: *Transportation Research Part A: Policy and Practice* 46.3, pp. 614–625. ISSN: 09658564. DOI: 10.1016/j.tra.2011.11.013. URL: https://linkinghub.elsevier.com/retrieve/pii/S0965856411001820 (visited on 01/07/2026).

Hu, Wen Xun and Amer Shalaby (Jan. 2017). "Use of Automated Vehicle Location Data for Route- and Segment-Level Analyses of Bus Route Reliability and Speed". en. In: *Transportation Research Record: Journal of the Transportation Research Board* 2649.1, pp. 9–19. ISSN: 0361-1981, 2169-4052. DOI: 10.3141/2649-02. URL: https://journals.sagepub.com/doi/10.3141/2649-02 (visited on 01/07/2026).

Shrestha, Sakar and Anil Marsani (2017). "Performance Improvement of a Signalized Intersection (A Case Study of New Baneshwor Intersection)". en. In.

A Appendix: Strategic Intersections and Color Code Visu

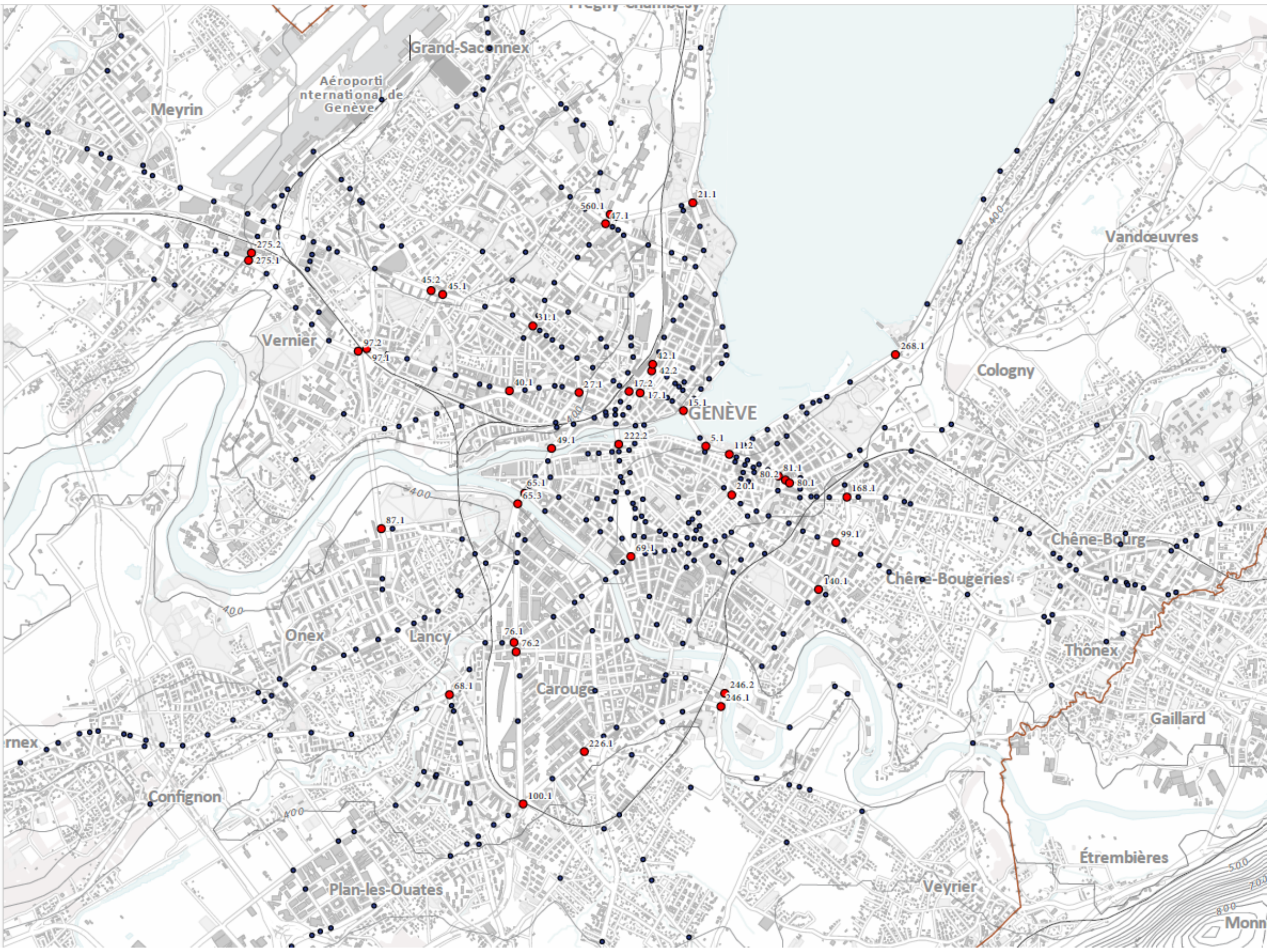


Figure 5: Map of the intersections in the Geneva urban network. In red the strategic intersections

Table 8: VISU color code - OCIT Standard

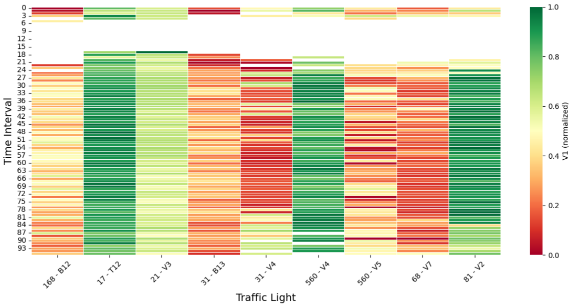
Color	**N°**	**Meaning**
Red Cli 1Hz	2	Transition time TC, towards free passage
Red	3	Stop
Red + Yellow	15	Transition time, towards free passage
Yellow	12	Transition time, towards stop
Green	48	Free passage
Green + Yellow Cli 1Hz	44	Free passage
Green + Yellow Cli 1Hz	56	Free passage
Green Cli 1Hz	32	For tricolor BàF: free passage
Green Cli 2Hz	16	For TC BàF: transition time, towards free stop
Green Cli 2Hz	96	Pedestrian, transition time towards stop
Green Cli 2Hz	98	Pedestrian, transition time towards stop
Yellow Cli 1Hz	8	Flashing

B Appendix: Description of Possible Patterns

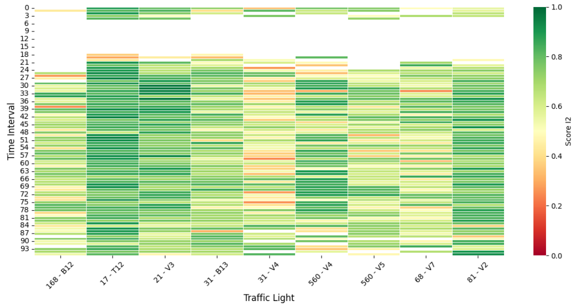
Table 9: Description and classification of the possible patterns within a time window

Pattern	Description	Classification
Always V	For the entire duration the color does not change and is always green. The vehicle arrives during the middle of a free passage phase	Randomness
Always R	For the entire duration the color does not change and is always red. The vehicle arrives in the middle of a red phase without being granted free passage	No priority
RV	During the time window the color changes from red to green.	Priority/Randomness
VR First	During the time window the color changes from green to red. The change occurs before the recording, the vehicle will have to wait at least half the duration of the time window before obtaining the right of way	No priority
VR After	During the time window the color changes from green to red. The change occurs after the recording, the vehicle will have at least half the duration of the time window to pass with the green light.	Priority/Randomness
RVR	The color changes twice during the time window. In this case the green phase will be short, suggesting that free passage was granted for the arrival of the vehicle.	Priority/Randomness
VRV	The color changes twice during the time window. In this case the red phase was short, suggesting that the vehicle was identified and free passage was granted even though the green phase had just ended	Priority
Other Patterns	All time windows with more than 2 color changes. Several changes in a few seconds suggest an adaptation of the signal based on the presence of the vehicle	Priority

C Appendix: Visualization of Performance Metrics

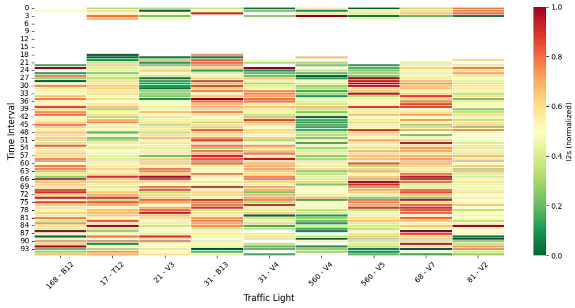


(a) V_1 - median speed difference



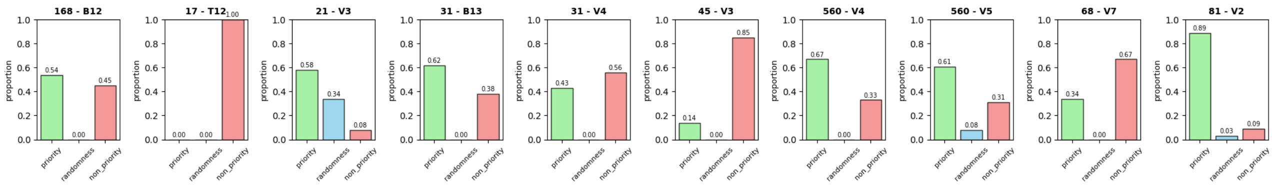
(b) I_1 - Crossing time score

(c) I_{2t} - Coefficient of variation of time

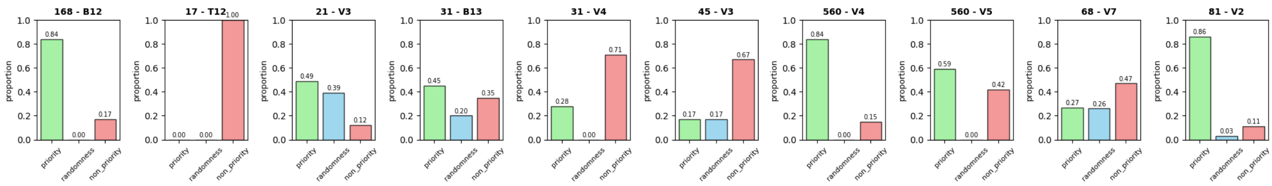


(d) I_{2s} - coefficient of variation of speed

Figure 6: Visualization of the performance metrics V1, I1 and I2



(a) Morning peak hours



(b) Evening peak hours

Figure 7: Visualization of the 3 categories defined according to the decision tree. I3 corresponds to the red column *Non priority*